



Project MNIST

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Motivation and introduction

- This project explores the cluster structure of the Ethiopic MNIST dataset, which contains grayscale images of handwritten Ge'ez numerals.
- This project demonstrates how I approach real-world data analysis. From raw data, to cleaning and transforming it, to exploring the structure, and generating insights.
- The goal wasn't to build a predictive model, but to analyze a complex dataset and understand its structure. I focused on comparing analytical approaches and understanding ambiguity in the data.

Dataset Overview

- 1000 Ethiopic handwritten digits.
- 28×28 grayscale images → 784 features/pixels + 1 column for label for what digit it is
- Pixel values normalized to [0,1]
- No missing values
- No traditional outliers (image data)
- Reshaped the image data into a tabular structure, and prepared it for analysis

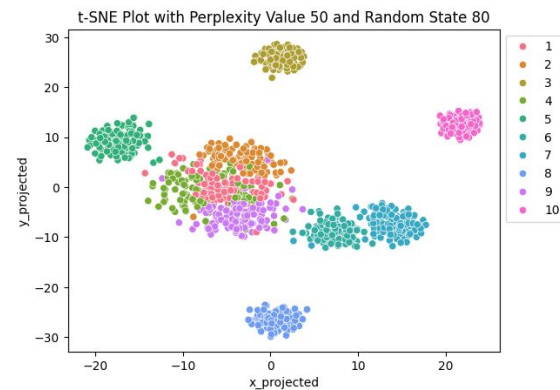
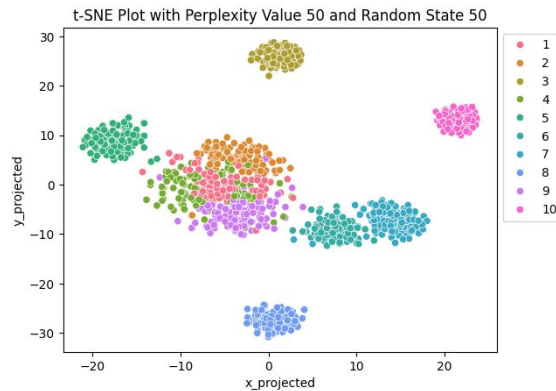
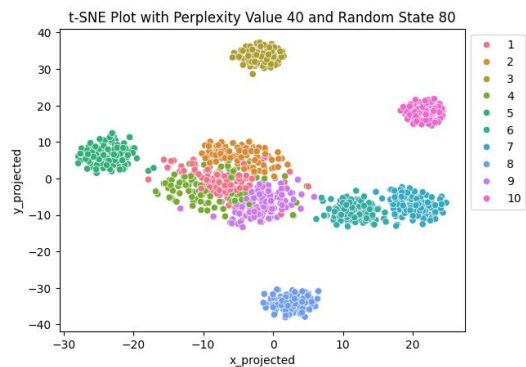
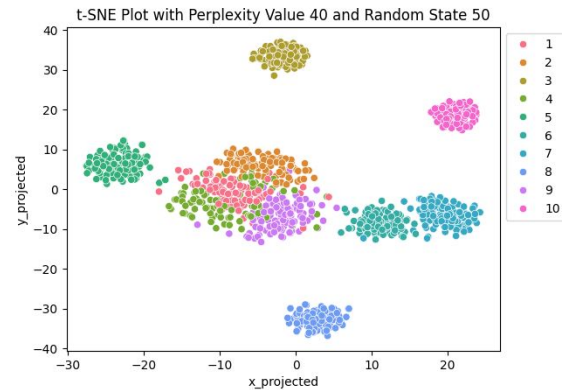
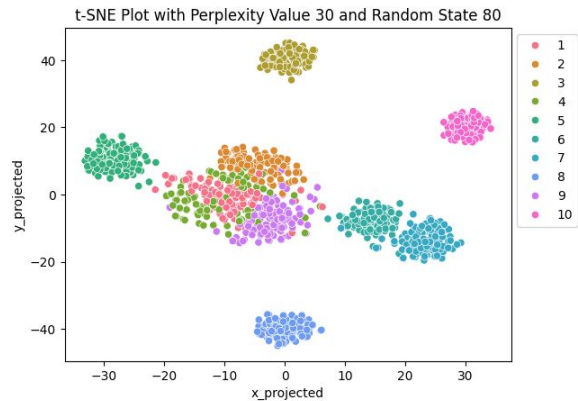
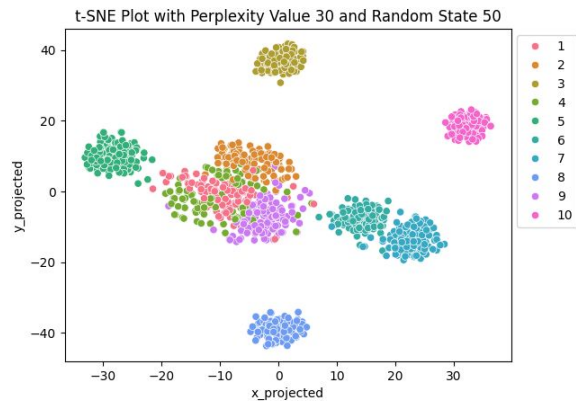
Averaged Image of Each Digit





Exploratory Data Analysis

- Before applying any methods, I explored the data visually and statistically to understand the variability, structure, and overlap between categories.
- This helped guide my analytical decisions rather than jumping straight into modeling
- I decided use t-SNE to visualize complex, high-dimensional data into a simpler 2D shape
- It making clusters and patterns visible by placing similar items close together and dissimilar items far apart



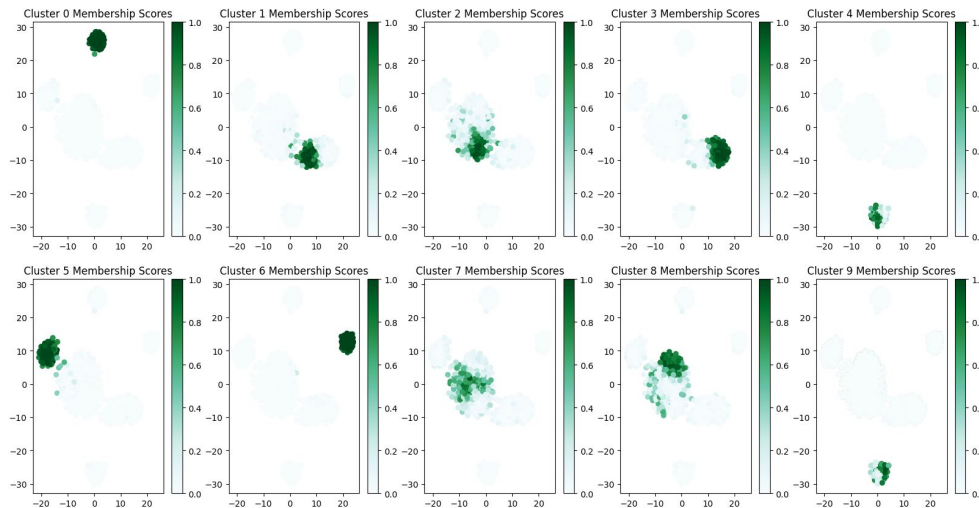


Choosing our clustering

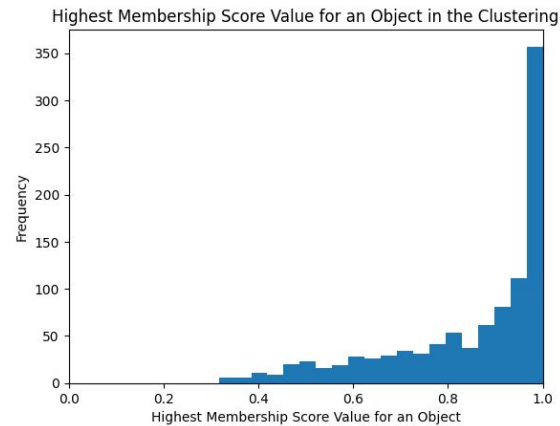
- t-SNE showed a large central cluster containing multiple digit labels, which suggests overlap in feature space. (Digits 1, 2, 4, and 9)
- Other digits were able to be cleanly separated
- A possible clustering algorithm to use could be Fuzzy C-means clustering
- Fuzzy C-Means allows each image to belong partially to multiple clusters, which better reflects uncertainty in ambiguous handwritten digits
- PCA to downsample the images and explore the relationship between the different digits, as there were some digits were very distinct and other were not as suggested by the T-SNE plots
- Lastly, since the clusters are mostly spherical, we also choose k-means as the alternative algorithm to compare against fuzzy c-means

Fuzzy C-means

Fuzzy Clustering: 10 clusters and $p=1.0942857142857143$



Images of the Cluster Centroids

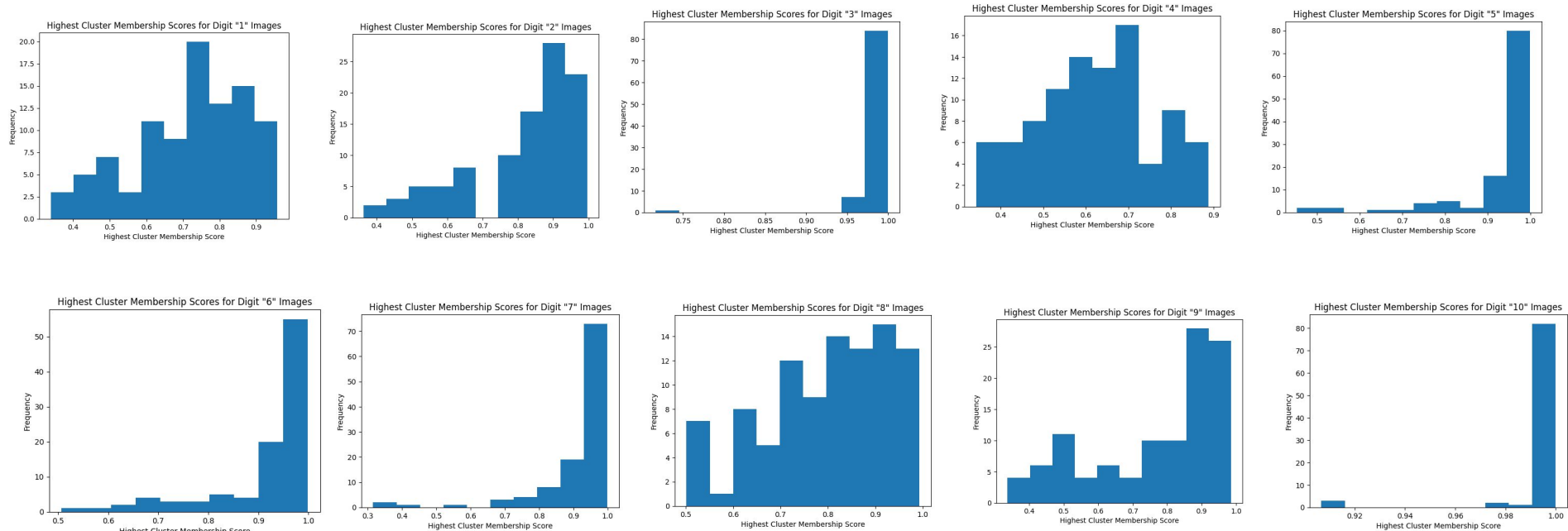


Low $p \rightarrow$ hard clustering

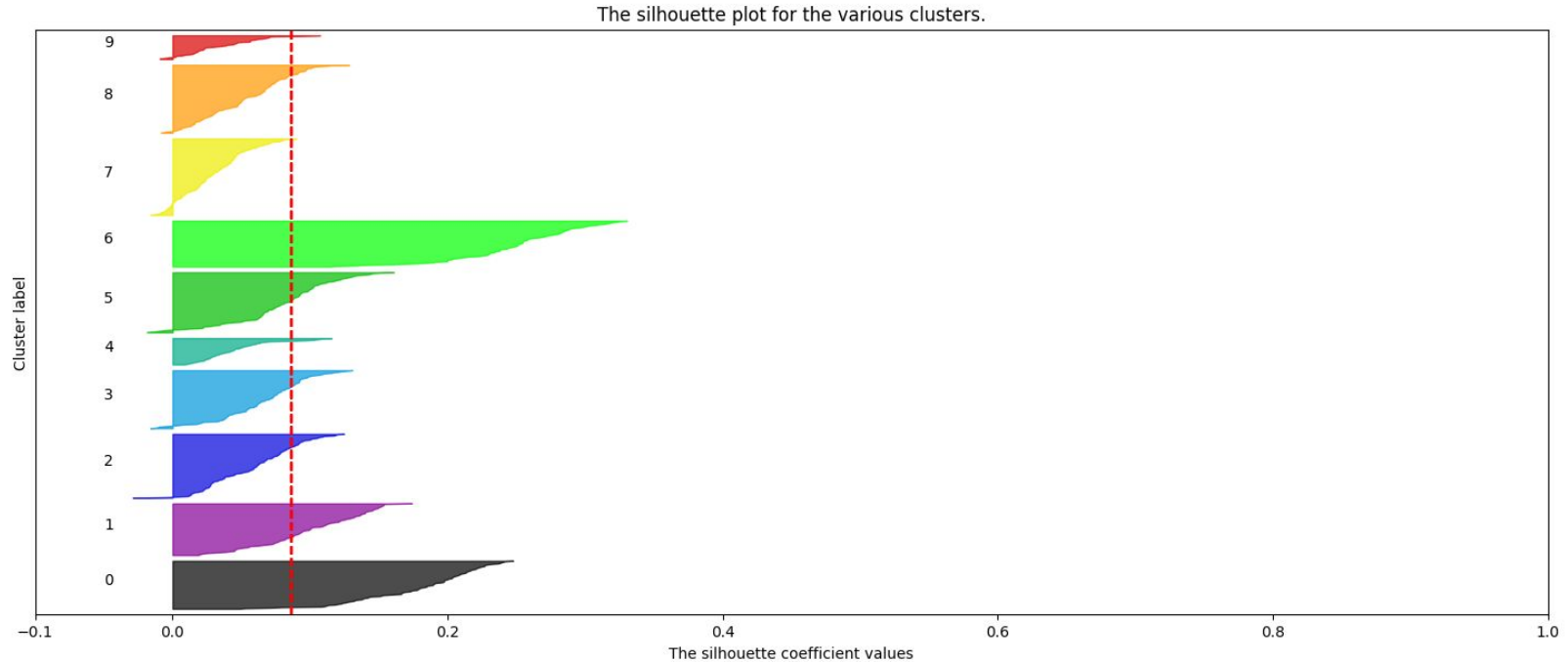
High $p \rightarrow$ very fuzzy

While we can gain some insight many of these values of p , we will choose to use the fuzzy clustering produced by $p=1.09$ as it allows us to classify most objects whose membership is relatively certain, reflect the uncertainty of the cluster assignment of "straddle nodes", and also determine more information about which clusters are fuzzier than others.

Membership Scores



Digit 1,2,4,8,9 have the least certainty in clustering, and 3,5,6,7,10 have most certainty in clustering.

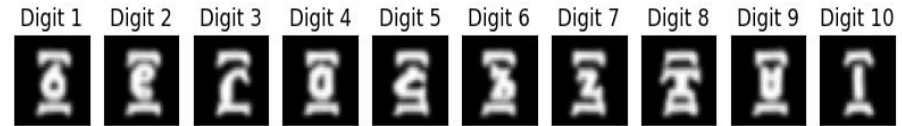


For $n_clusters = 10$ The average silhouette_score is : 0.086, and adjusted rand score= 0.7735
Low silhouette score is expected here because silhouette assumes hard assignments and convex clusters. Since handwritten digits overlap in feature space, especially for similar glyphs, the silhouette metric underestimates the quality of fuzzy clustering.

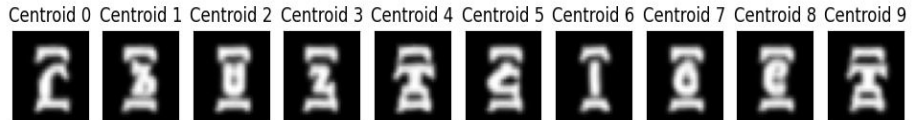
Results

- The results from the Fuzzy C-Means clustering of a handwritten digit dataset highlight that most clusters correspond well to specific digits, demonstrating the effectiveness of this clustering approach.
- Clusters 0, 1, 2, 3, 5, 6, 8 are well-defined.
- However, Cluster 7 presents some ambiguity; its centroid appears to represent a blend of digits 1 and 4. This observation is supported by findings from our t-SNE plot, which also indicates overlapping features between these two digits in this particular cluster.
- Cluster 4 and 9 both have centroids that resemble a digit 8.

Averaged Image of Each Digit

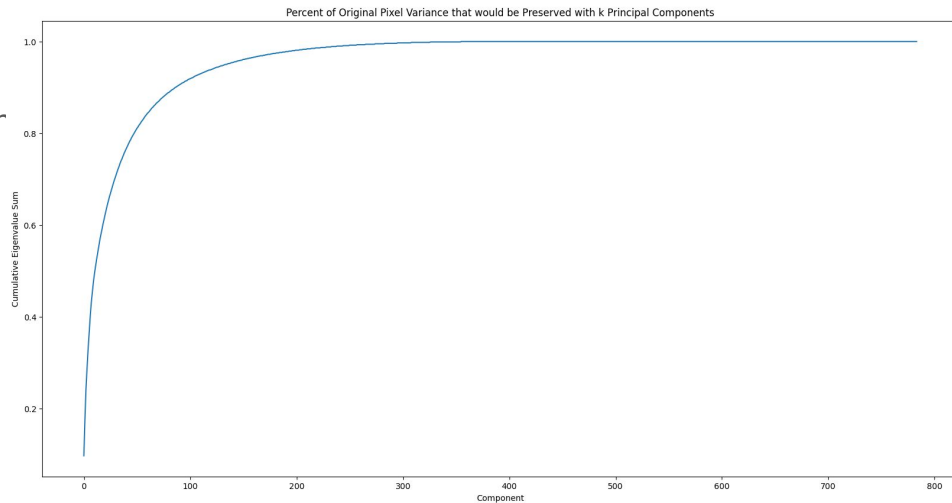


Images of the Cluster Centroids



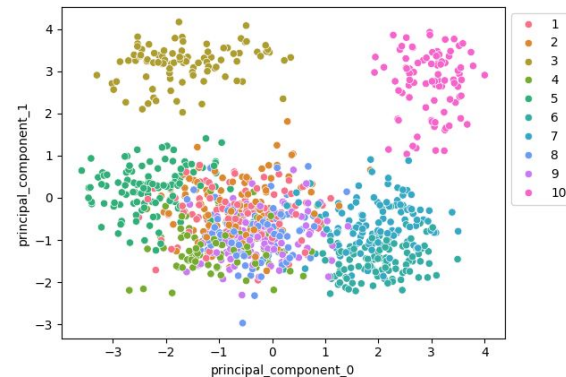
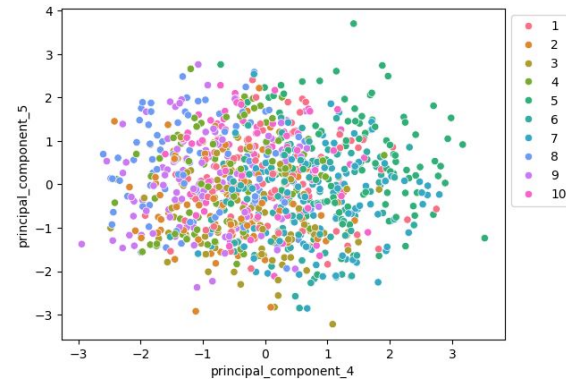
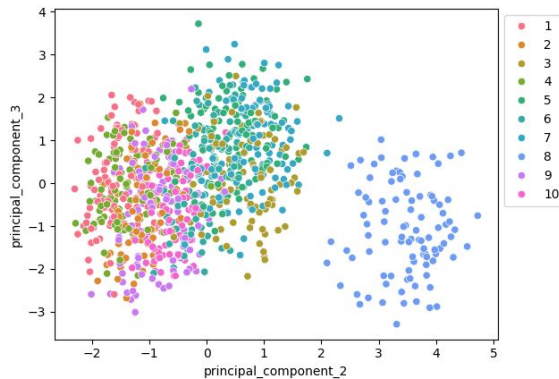
PCA

- Downsample the images
- Explore the relationship between each of the digits in the components
- 30-PCs will capture 70% of the original variation
- Making K-means more stable and interpretable.



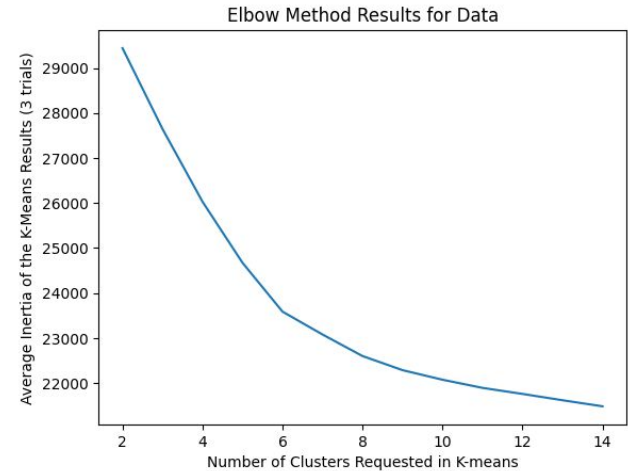
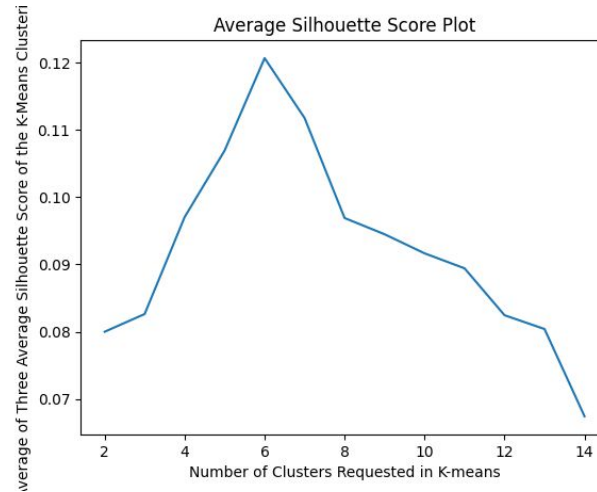
PCA Plots

- Clear distinction between Digit 3, 8, 10, and the other digits
- First few PCs capture major stroke patterns
- No structures in the PC4 and PC5 plots
- PCA is linear, so it won't fully separate nonlinear structures



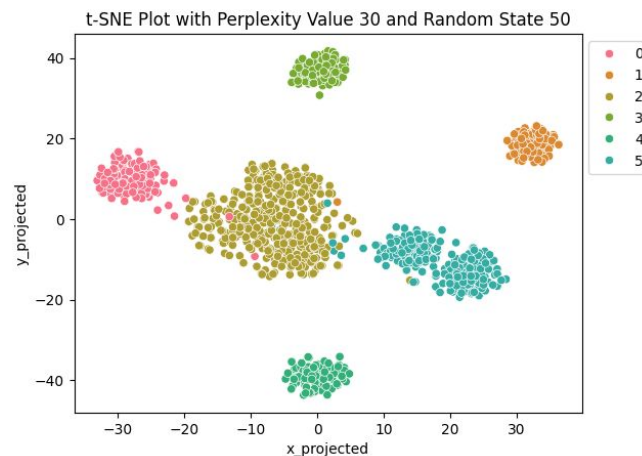
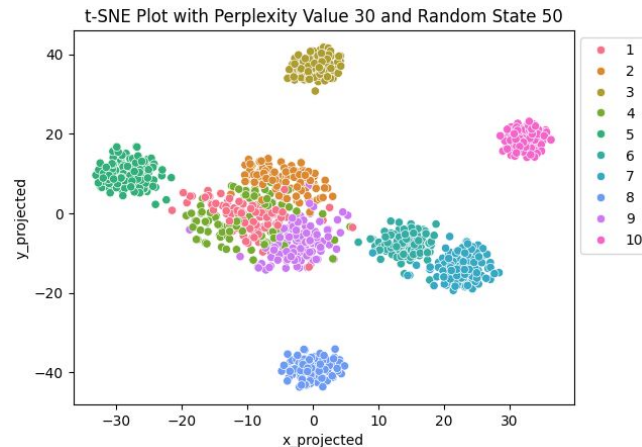
K Means

- We will use both the Elbow and the Silhouette methods to find the optimal K.
- Both methods suggest $k=6$



Clustering Results

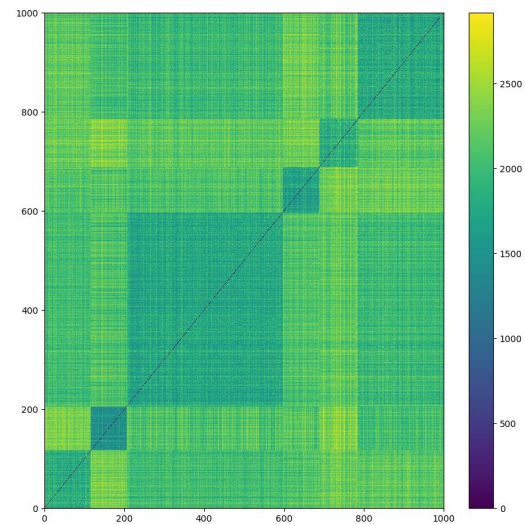
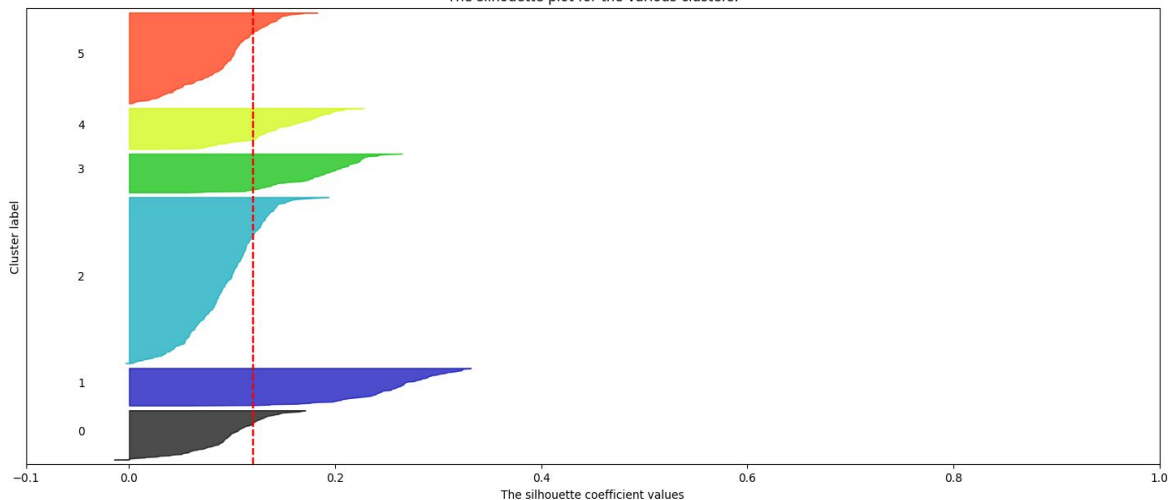
- K-means was able to locate the clusters with good separation.
- However it merged every Digits that had poor distinct separation to another digit. Comparing the T-SNE plots, we see the very distinct digits were classified correctly,
- There are only a few mis-identification between Cluster 0 (Digit 5) and Cluster 2. Cluster 5 (Digits 6 and 7) also had some but overall k-means was able to find the inherent structure as long as it is not too similar.



Cluster Exploration

- The average silhouette_score is : 0.12,
- Adjusted rand score = 0.51

The silhouette plot for the various clusters.





Discussion

- K-means was only able to detect very distinct clusters
- K-mean clusters have poor within cluster similarity and cohesion
- K Means rand score = 0.51 and Fuzzy c-mean rand score = 0.77
- Kmeans shows that there were 6 groupings while Fuzzy was able to detect 10
- Fuzzy can detect the relationship between single data point to multiple clusters, but kmeans can not.



Conclusion

Summary Both Fuzzy C means and K-means were successful in finding useful clustering solution, PCA was successful in a image downsample role

Recommendation We would recommend using Fuzzy c-means to find similarities between the digits. Especially the cluster membership scores as they are a good indicator

Shortcomings K-means had a difficult time distinguishing more than 6 groups.



How This Relates to the Role

- This project reflects the core responsibilities of this role: preparing data, analyzing it to find patterns, and communicating insights that help guide decisions.
- I primarily used Python for data manipulation, analysis, and visualization. Following a workflow similar to analytics pipelines.
- Something I can improve would be, to store transformed data and analytical outputs in a relational database and use SQL to query results for reporting or dashboards.
- Overall, this project shows how I approach data problems methodically.



References

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